higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

MECHANOTECHNICS N6

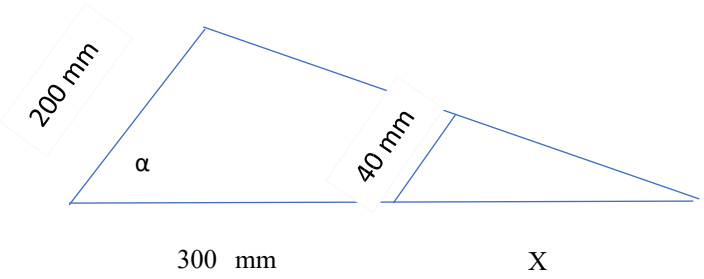
21 NOVEMBER 2019

This marking guideline consists of 11 pages.

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QUESTION 1

1.1



$$\frac{200}{40}=\frac{300+x}{x}$$

$$5x = 300 + x$$

$$x = 75 \text{ mm}$$

$$OE = 300 + 75 = 375 \ mm$$

$$\cos\alpha = \frac{200}{375}$$

$$\alpha = 57{,}77^{o}$$

$$\therefore \Theta = 270 - 57{,}77$$

$$\Theta = 212{,}23^{o}$$

(5)

1.2 $$P = \frac{2\pi NT}{60}$$

$$5 \times 10^{3} = \frac{2\pi(450)T}{60}$$

$$\therefore T = 106{,}103 \ Nm$$

(2)

1.3 $\frac{T_1}{T_2} = e^{\mu\theta}$

$\frac{T_1}{T_2} = e^{(0,25)(3,704)}$

$T_1 = 2,524\ T_2$

$T = (T_1 - T_2)R$

$106,103 = (2,524\ T_2 - T_2)\ 0,2$

$T_2 = 348,107\ N$

$T_1 = 878,622\ N$

$T_2 \times 40 = F \times 450$

$348,107 \times 40 = F \times 450$

$F = 30,943\ N$ (7)

[14]

QUESTION 2

2.1 $R_m = \frac{2}{3}\left(\frac{R^3 - r^3}{R^2 - r^2}\right)$

$= \frac{2}{3}\left(\frac{(0,1)^3 - (0,05)^3}{(0,1)^2 - (0,05)^2}\right)$

$= 77,778\ mm$

$P = \frac{2\pi NT}{60}$

$20 \times 10^3 = \frac{2\pi(1\ 000)T}{60}$

$T = 190,986$

$T = \mu F_A \times R_m \times n$

$190,986 = 0,4\ F_A \times 77,778 \times 10^{-3} \times 6$

(6)

$F_A = 1\ 023{,}136\ \ N$

2.2 $F_{(reduction \in spring\ \ force)} = F_A - F_{(New\ \ force\ \ after\ \ wear)}$

$= 1\ 023{,}136 - 632{,}136$

$= 391\ \ N$

$x_{(total\ \ wear)} = number\ \ of\ \ contact\ \ surfaces \times wear\ \ on\ \ each\ \ contact$

$= 12 \times 0{,}8$

$= 9{,}6\ \ mm$

$$spring\ \ stiffness = \frac{F_{(reduction \in spring\ \ force)}}{x_{(total\ \ wear)} \times n_{(number\ \ of\ \ spring\ \ pair\ \ discs)}}$$

$$= \frac{391}{(9{,}6)(6)} = 6{,}788\ \ N/mm$$

(6)
[12]

QUESTION 3

3.1 $$P = \frac{2\pi NT}{60}$$

$$50\ \ x\ \ 10^3 = \frac{2\pi(900)T_A}{60}$$

$$T_A = 530{,}156\ \ Nm$$

(2)

3.2 $D_A N_A = D_B N_B$

$(150)(900) = (280)N_B$

$N_B = 482{,}143\ \ r/min$

$$P = \frac{2\pi N\, T_B}{60}$$

$$50\ \ x\ \ 10^3 = \frac{2\pi(482{,}143)T_B}{60}$$

$T_B = 990{,}297\ \ Nm$

(4)

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3.3 $$F_t = \frac{T \times 2}{PC\ D_A} = \frac{(530,156)(2)}{0,15} = 7,069\ \ kN$$ (1)

3.4 $$F_R = F_t \tan \phi = (7\ \ 069) \tan 20° = 2,573\ \ kN$$ (1)

3.5 $$\frac{T_1}{T_2} = e^{\mu\theta}$$

$$\frac{T_1}{T_2} = e^{(0,4)(\pi)}$$

$$T_1 = 3,514\, T_2$$

$$T_B = (T_1 - T_2)$$

$$990,297 = (3,514\, T_2 - T_2)0,2$$

$$T_2 = 1\ \ 969,564\ \ N$$

$$T_1 = 6\ \ 921,049\ \ N$$ (5)

3.6 $$\Sigma CWM = \Sigma ACWM$$

$$(R_R \times 1,2) + (2\ \ 573\ \ \times 0,7) = (810\ \ \times 0,35) + (1\ \ 969,564 + 6\ \ 921,049)1,4$$

$$R_R = 9\ \ 107,715\ \ N$$

$$\Sigma CWM = \Sigma ACWM$$

$$(R_L \times 1,2) + (1\ \ 969,564 + 6\ \ 921,049)0,2 + (2\ \ 573 \times 0,5) = 810 \times 0,85$$

$$R_R = \text{-}1\ \ 980,102\ \ N$$ (4)

[17]

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QUESTION 4

4.1 $V_{shaft} = Al = \pi\left(0{,}025^2 - 0{,}015^2\right) \times 0{,}4 = 502{,}655 \times 10^{-6}\, m^3$

$I_{shaft} = \rho v \times \frac{1}{2}\left(R^2 + r^2\right)$

$= (7\ \ 800)\left(502{,}655 \times 10^{-6}\right) \times \frac{1}{2}\left(0{,}025^2 + 0{,}015^2\right)$

$= 1{,}666 \times 10^{-3}\ \ kg.m^2$

$I_{flywheel} = mk^2$

$= 500 \times 0{,}3^2$

$= 45\ \ kg.m^2$

$I_{total} = I_{shaft} + I_{flywheel}$

$= 46{,}666\ \ kg.m^2$ (7)

4.2 $E_k = \frac{1}{2} Iw^2$

$¿ \frac{1}{2}(46{,}666)\left(\frac{2\pi}{60} \times 250\right)^2$

$= 15{,}992\ \ kJ$ (2)

4.3 $V = wr$

$¿ \frac{2\pi}{60} \times 250 \times 1{,}2$

$= 31{,}416\ \ m/s$ (2)

[11]

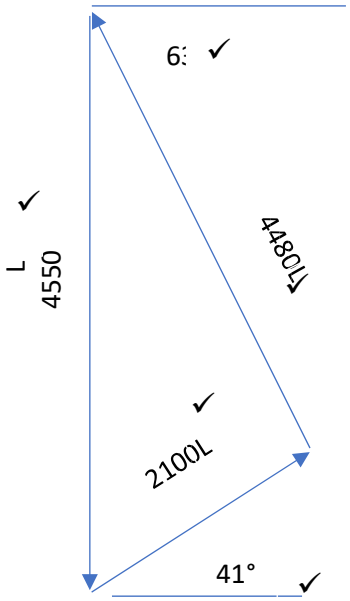
QUESTION 5

5.1

	m(kg)	r (mm)	mr(kg.mm)	l ¿	$mr\,l$ ¿
A	5	300	1 500	$3l$	4 500 l
B	7	250	1 750	$2l$	3 500 l
C	6	350	2 100	l	2 100 l
D	m	400	400 m	0	0

(4)

(Each tick in the table = $\frac{1}{2}$ mark) (8 × 1/2)

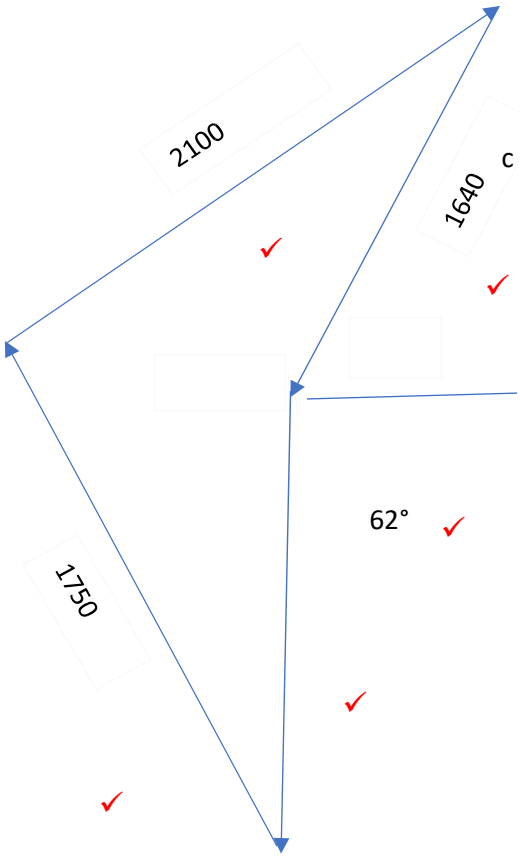


Angle between A and B is 153 °
Angle between A and C is 131 °

(7)

5.2 $1\ \ 640 = 400\ \ mm$

$m = 4.1\ \ kg$



(6)
[17]

QUESTION 6

6.1 $T_E = M_L \times 1\ 000 \times \mu \times g \times \%\ m$

$= 40 \times 1\ 000 \times 0{,}25 \times 9{,}81 \times 0{,}7$

$= 68\ 670\ N$ (2)

6.2 $T_E = F_{RL} + F_{RT} + F_a + F_g$

$F_R = R \times m_L = 45 \times 40 = 1\ 800\ N$

$F_R = R \times m_T = 40 \times 400 = 16\ 000\ N$

$F_g = W \sin\theta = \left[(400 + 40)1\ 000\right]\frac{1}{50} = 8\ 800\ N$

$68\ 670 = 1\ 800 + 16\ 000 + F_a + 8\ 800$

$F_a = 42\ 070\ N$

$F_a = ma$

$42\ 070 = (440 \times 1\ 000)a$

$a = 0{,}956\ m/s^2$

$v = u + at$

$v = \frac{50}{3{,}6} = 13{,}889\ m/s$

$13{,}889 = 0 + (0{,}956)t$

$t = 14{,}528\ sec$ (9)

6.3 $s = ut + \frac{1}{2}at^2$

$s = 0 + \frac{1}{2}(0{,}956)(14{,}528)^2$

$s = 100{,}888 \ \ m$

$P_{LOCO} = T_E(\frac{s}{t})$

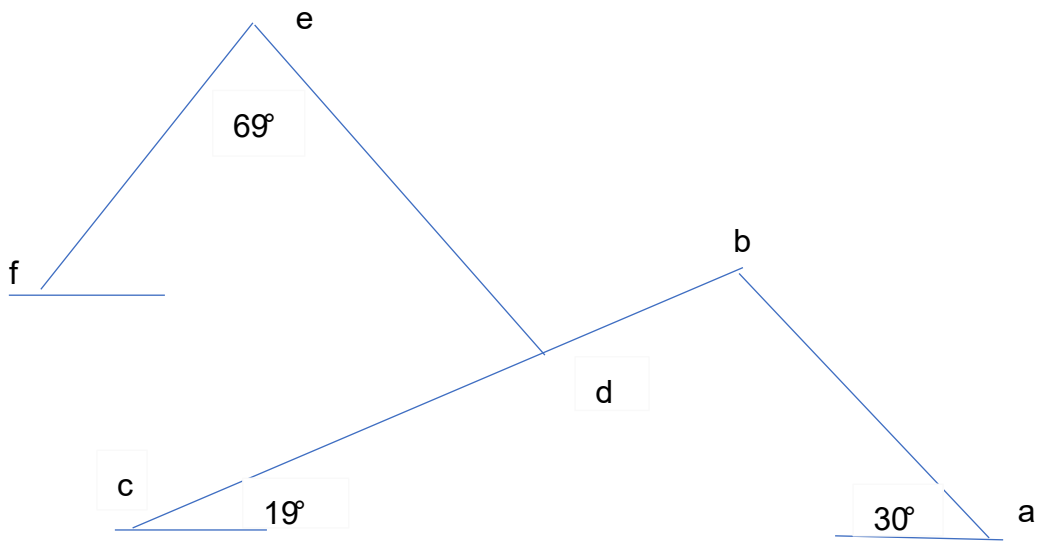
$= (68\ \ 670)(\frac{100{,}888}{14{,}528})$

$= 476{,}87 \ \ kW$ (4)

[15]

QUESTION 7

7.1



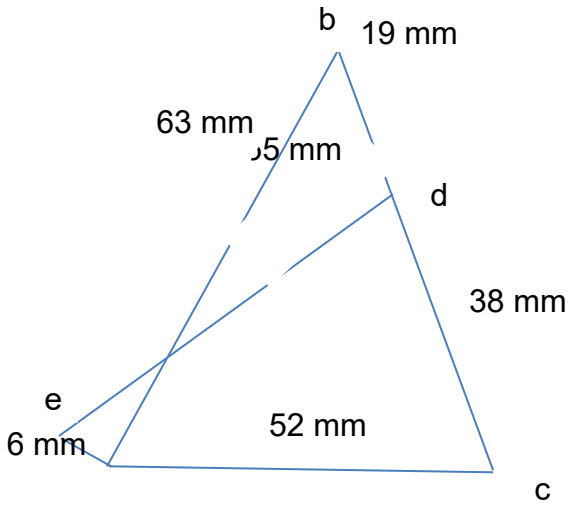
(4)

7.2 $v = \left(\frac{2\pi N}{60}\right)r$

$¿\left(\frac{2\pi(150)}{60}\right)0{,}1$

$= 1{,}571 \ \ m/s$ (2)

7.3



(6)

7.4 $V = 52 \div 40 \ = 1{,}3 \ \ m/s$ (2)

[14]

TOTAL: 100